UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Fall 2023

Paper: Computational Physics-II
Course Code: PHYS-3504

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answers of the following Questions in MATLAB. (6x5=30)

i) To generate an array R. Find out zero & nonzero elements and index numbers of R. Find maximum value and index number of R.
ii) Write example to show the function of the following.
(a) min() (b) zeros() (c) size() (d) roots() and (e) diff()
iii) Write syntax with example for the following in MATLAB:
(a) for() loop (b) ginput() and (c) gtext()
iv) Create a vector H with values 100 to-1 having decrement 2. Find sum of H, plot H and print H in reverse order.
v) To calculate by a function the area and circumference of a circle by taking the values of radius from the user
vi) Write one example to show the use of: poly(), polyfit() & rand()

Answer the following questions. (3x10=30)

Q.2.	How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of an object for 31 collisions. Plot estimate graph. Write a program to find out factor of a number. Write MATLAB program to evaluate $\int_1^6 (3\sqrt{x}-0.5)dx$.
Q.3.	Suppose A be a 3x3 matrix. Write MATLAB program which reads in random numbers as entries of the matrix A and calculate (i) sum and average of the all matrix elements, (ii) transpose of the matrix A (also plot the matrix), (iii) also check whether the Matrix A is an identity matrix? (iv) sort the matrix elements, (v) divide matrix rows by its row average. Write program to study the damped harmonic motion (DHM) of a mass attached with a spring using Euler's method with: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and tmax 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms,). Print and plot values for time, position, velocity and acceleration. Also suggest changes to write program for Simple harmonic motion.
Q.4.	Write MATLAB program to plot and print time, position, velocity and acceleration values for a spherical object in air drag. The acceleration and other parameters are given by $a = g - k v^2$ where $k = c \pi \rho r^2 / 2m$ with the conditions: g = 9.8 m/sec^2, c=0.46 (drag constant), ρ= 1.2 kg/m^3, r=1m, v=0 m/sec, h= 0.1 sec and tmax=2.5sec. Write code to solve for the system of equations for x, y and z by using two different methods.Such that 2x+y+2z =17, 4x-3y+8z=6 and x-3y+z=6.

UNIVERSITY OF THE PUNJAB

Roll No. 9852

Associate Degree : Sixth Semester – Fall 2023

Course Code: PHYS-3504

Time: 3 Hrs.

ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVID

Write short answers of the following Questions. (6x5=30)

i) Write one example to show the use of: roots(), diff(), rand(), and polyval()

ii) Create a vector y with values 1 to 100 having increment 2. Find sum of y, plot y and print y in reverse order.

iii) Give example to generate and print values of two vectors x & y. Also draw graphs of x vs y using stem(), bar() and plot().

5 iv) To calculate by a function the area and circumference of a circle by taking the values of radius from the user

v) ~~If t =[0 2π], s1=sin(x) & s2= cos(x),~~ Write code to plot t vs s1, t vs sum of s1 & s2, t vs mean of s1 & s2 and find maximum of s2.

2 vi) Write example to show the function of the following.
(a) max() (b) ones() (c) length() (d) mean() and (e) std()

Answer the following questions. (3x10=30)

Q.2. Write program to plot x vs polynomial $3x^3 - 7x^2 + 2x + 4$ for [-15 15] using two different methods, the interval is given by: h= 0.01. Also, print values of *x* and *f(x)*, find out sum, minimum and average of *f(x)* values. Sort data of *f(x)* in descending order.

Write MATLAB program to plot time (t) against y1=sin(2t) and y2=cos(4t) curves on the same figure. Use t = [0 4π]

Write MATLAB program to determine equivalent capacitor of n capacitors connected in parallel.

Q.3. Write MATLAB program to plot and print time, position, velocity 6 and acceleration values for a spherical object in air drag. The acceleration and other parameters are given by **$a = g - k v^2$** where **$k = c \pi \rho r^2 / 2m$** with the conditions: **$g = 9.8$ m/sec^2**, **c=0.46** (drag constant), **ρ= 1.2 kg/m3, r=1m, v=0 m/sec, h= 0.1 sec** and **tmax=2.5sec.**

Write general syntax for a user defined function in MATLAB. Write a program to implement a function **circle** to calculate area and circumference of a circle for n such that n is [2.5:0.5: 4.5].

Q.4. How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of an object for 31 collisions. Plot estimate graph.

Write a program to find out factor of a number.

Write a program to evaluate $\int_0^{\pi} \sin(x)dx$.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2023

Paper: Computational Physics-II   Course Code: PHYS-3504   Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (6x5=30)

i) Give example to generate and print values of two vectors x & y. Also draw graphs of x vs y using stem(), bar() and plot().

ii) Write example to show the function of the following.
 (a) max() (b) ones() (c) length() (d) mean() and (e) std()

iii) If t =[0 2π], s1=sin(x) & s2= cos(x), Write code to plot t vs s1, t vs sum of s1 & s2, t vs mean of s1 & s2 and find maximum of s2.

iv) Create a vector y with values 1 to 100 having increment 2. Find sum of y, plot y and print y in reverse order.

v) To calculate by a function the area and circumference of a circle by taking the values of radius from the user

vi) Write one example to show the use of: roots(), diff(), rand(), and polyval()

Answer the following questions. (3x10=30)

Q.2. How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of an object for 31 collisions. Plot estimate graph.
Write a program to find out factor of a number.

Write a program to evaluate $\int_0^{\pi} \sin(x)dx$.

Q.3. Write program to plot x vs polynomial $3x^3 - 7x^2 + 2x + 4$ for [-15 15] using two different methods, the interval is given by: h= 0.01. Also, print values of *x* and *f(x)*, find out sum, minimum and average of *f(x)* values. Sort data of *f(x)* in descending order.

Write MATLAB program to plot time (t) against y1=sin(2t) and y2=cos(4t) curves on the same figure. Use t = [0 4π]
Write MATLAB program to determine equivalent capacitor of n capacitors connected in parallel.

Q.4. Write MATLAB program to plot and print time, position, velocity and acceleration values for a spherical object in air drag. The acceleration and other parameters are given by **a = g – k v²** where **k = c π ρ r² / 2 m** with the conditions: **g = 9.8 m/sec²**, **c=0.46** (drag constant), **ρ= 1.2 kg/m3, r=1m, v=0 m/sec, h= 0.1 sec** and **tmax=2.5sec.**

Write general syntax for a user defined function in MATLAB. Write a program to implement a function circle to calculate area and circumference of a circle for n such that n is [2.5:0.5: 4.5].

UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Sixth Semester - Spring 2023

Roll No. ....

Paper: Computational Physics-II   Course Code: PHYS-3504   Time: 3 Hrs.   Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: **(6x5=30)**

I. Explain with example the following terms in MTLAB:

a. polyval()

b. roots()

c. factor()

II. Write syntax with example for the followings in MATLAB:

a. Extrapolation of data

b. Curve fitting

III. Write a MATLAB program with function to calculate $f(x)$, given by:

$$f(x) = 28x^3 - 15x^2 + 7x - 3$$

Calculate, print and plot values for x and $f(x)$. Also find minimum and maximum of $f(x)$ values for fifteen different values of x.

IV. Write MATLAB to plot 'x' against f(x) values using supbplot such that: $x = [0\ 4\pi]$, $y(x) = n*\sin(x)$ and n=1,2,3,4

V. Write MATLAB program to determine corresponding force values for the work and distance values as give in the table. Also calculate total force and average work values.

w (work)	20 27 47.5 62
d(distance)	12 27 41 56

VI. Two vectors are given:

u= 5i+9j-2k and v= 4i-6j+12k

Use MATLAB to calculate the dot product u.v of the vectors in two ways:

a) Define u as row vector and v as a column vector, and then use matrix multiplication.

b) Use the dot function.

Solve the following.

Q.2. (a) Write a MATLAB program to plot x vs ψ such that:

$$\psi_n(x) = A\frac{n\pi x}{L}$$

Where $A = \sqrt{\frac{2}{L}}$, L = 4, n = [1:4] **(6 Marks)**

(b) Write MATLAB program to evaluate the $\int_1^4 \left(\frac{\sqrt{x}}{6}\right)dx$ by Trapezoidal Rule. (n=6) **(4 Marks)**

Q.3. Write MATLAB program to describe motion of a charged particles in magnetic field. Also draw estimate output graphs with proper curve labels, x & y labels and title. **(10 Marks)**

Q.4. Create matrix D in MATLAB such that

$$D = \begin{bmatrix} 10 & 9 & 8 & 7 & 6 \\ 12 & 14 & 16 & 18 & 20 \\ 1 & 4 & 7 & 10 & 13 \\ 25 & 8 & 11 & 2 & 1 \\ 33 & 14 & 29 & 6 & 12 \end{bmatrix}$$

(i) Write v1 equals 4^{th} column of D. (ii) find location of value 18 in v1, (iii) create a linear vector k containing all elements of D. (iv) create v2 equals 3^{rd} row of D. (v) determine u= v1.v2 (vi) plot 4^{th} column against 1^{st} row of D (vii) select and sort in [illegible]ending order the 4^{th} and 5^{th} rows of D. (viii) select a 3x3 middle matrix as m out of D. (10

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2023

Roll No. ....

Paper: Computational Physics-II   Course Code: PHY-422   Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following. **(6x5=30)**

i) Give an example to generate and display elements of a Matrix A. Also find out maximum, standard deviation, sum and mean of A.

ii) Write example to show the function of the following.
(a) cumprod() (b) pretty() (c) solve() (d) length() and (e) sum()

iii) If x =[1 2 3] y1 =[16 22 14] & y2 = [13 22 21]. Write code to plot x vs y1, x vs sum of y1 & y2, x vs mean of y1 & y2 and find transpose of y1.

iv) If x = [3 6 5 8 4] & y = [3 5 6 7 4]. Write code to find values of x, number of times and locations when x is less than y.

v) Write code to general any two polynomials f1 and f2. Find sum, difference and division of f1 and f2.

vi) Write one example to show the use of the following: while. polyfit(), rand(), and diff()

Solve the following. **(3x10=30)**

Q.2. Write MATLAB code to simulate Brownian motion. Also calculate the total distance and average distance traced by a particle during Brownian motion describe by (x,y) values.
Write code to solve for the system of equations for x, y and z by using two different methods.Such that 2x+y+2z =17, 4x-3y+8z=6 and x-3y+z=6.

Q.3. Write a program for full wave rectifier. Also draw input and output graphs with proper labels.
The temperature measurement of containers A and B is given in the table. Write program to determine: (i) temperature of A greater than that of B (ii) temperature of B equal to that of A with day (iii) temp. of B > 60 with days and (iv) find out the day when average temperature of A and B is maximum.
Write

Day	1	2	3	4	5
A	60	56.2	63	64	58
B	54.6	56.2	64	60	59

Q.4. Write MATLAB program to study the damped harmonic motion (DHM) of a mass attached with a spring using Euler's method with: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and tmax 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms,). Print and plot values for time, position, velocity and acceleration. Also write program for Simple harmonic motion.
Write code to plot graph of t against y(t) such that $y(t) = e^{-8t}\sin(9.7t + \pi/2)$ and 0<t<2pi. Also plot differential of y. Find median of y.

UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Sixth Semester – Spring 2022

Roll .....

Paper: Computational Physics-II   Course Code: PHYS-3504   Time: 3 Hrs.   Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Write short answer of the following questions: (6x5=30)

I. Explain with example the following terms in MTLAB:
   a. dsort()
   b. diff()
   c. disp()

II. Write syntax with example for the followings in MATLAB:
   a. 2D interpolation
   b. Curve fitting

III. Calculate the radius r of a sphere that has a volume of $350 m^3$. Once r is determined, use this r to calculate surface area of sphere using MATLAB program.

IV. Write MATLAB program to plot time (t) against sin(t), and 5cos(t) curves on the same graph with different colors. Use $t = [0\ 4\pi]$

V. Two vectors are given:
u= 6i+4j-9k and v= -3i+4j-8k
Use MATLAB to calculate the dot product u.v of the vectors in two ways:
a) Define u as row vector and v as a column vector, and then use matrix multiplication.
b) Use the dot function.

VI. Write MATLAB program to calculate time period of simple pendulum ($T=2\pi\sqrt{\frac{l}{g}}$) using functions. Ask user to input value of *l*.

Solve the following.

Q.2. (a) Write a MATLAB program to plot x vs ψ such that:

$$\psi_n(x) = A\frac{n\pi x}{L}$$

Where $A = \sqrt{\frac{2}{L}}$, L = 4, n = [1:4] **(6 Marks)**

(b) Write MATLAB program to evaluate the $\int_1^4 (\frac{\sqrt{x}}{6})dx$ by Trapezoidal Rule. (n=6) **(4 Marks)**

Q.3. The acceleration of a spherical body experiencing air drag is given by $a = g - k v^2$ where $k = c\pi \rho r^2 / 2m$ with the conditions: $g = 9.8$ m/sec², c=0.46 (drag constant), ρ= 1.2 kg/m³, r=1m, v=0 m/sec, h= 0.1 sec and tmax=2.5sec. Write MATLAB program to plot and print time, position, velocity and acceleration values. Also draw estimate output graphs with proper curve labels, x & y labels and title. **(10 Marks)**

Q.4. What are random numbers? Write syntax for random number generation in MATLAB. Write MATLAB program to plot graph for a particle describing Brownian motion. Also calculate total and average distances traced by the particle. Draw approximate output graph with proper labels. **(10 Marks)**

Roll No. 00.38.83

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – Spring 2022

Paper: Computational Physics-II | Course Code: PHY-422 | Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Solve the following questions. **(6x5=30)**

i) Give an example to generate and display elements of an array. Also find out maximum, standard deviation, sum and mean of data.

ii) Write example to show the function of the following.
(a) prod() (b) min() (c) int() (d) size() and (e) diag()

iii) If x =[1 2 3] y1 =[16 22 14] & y2 = [13 22 21], Write code to plot x vs y1, x vs sum of y1 & y2, x vs mean of y1 & y2 and find transpose of y1.

iv) If x = [16 5 13 25 15] & y = [7 5 20 22 10]. Write code to find values of x, number of times and locations when x is greater than y.

v) Write code to general any two polynomials f1 and f2. Find sum, difference and product of f1 and f2.

vi) Write one example to show the use of the following: for, polyval(), rand(), and diff()

Solve the following questions. **(3x10=30)**

Q.2. Write MATLAB code to simulate Brownian motion. Also calculate the total distance and average distance traced by a particle during Brownian motion describe by (x,y) values.

Write code to solve for the system of equations for x, y and z by using two different methods.Such that $2x+y+2z=17$, $4x-3y+8z=6$ and $x-3y+z=6$.

Q.3. Write general syntax for a user defined function in MATLAB. Write a program to implement the functions volume and power such that volume = $\pi^2 r h$. and power = $(2\pi N T)/60$. Take values of variables from the user.

If y =[45 0 32 0 89 0 44]. Write program to find out number of zero values with their index numbers, number of nonzero values, and sum of nonzero values of y. Find maximum of y. Also plot y.

Q.4. The acceleration of a spherical body experiencing air drag is given by $a = g - k v^2$ where $k = c \pi \rho r^2 / 2m$ with the conditions: g = 9.8 m/sec^2, c=0.46 (drag constant), ρ= 1.2 kg/m3, r=1m, v=0 m/sec, h= 0.1 sec and tmax=2.5sec. Write MATLAB program to plot and print time, position, velocity and acceleration values. Also draw estimate output graphs with proper curve labels, x & y labels and title.

Write code to plot graph of t against y(t) such that $y(t) = e^{-8t} \sin(9.7t + \pi/2)$ and 0<t<2pi. Also plot differential of y. Find median of y.

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – 2021

Roll No. 02565

Paper: Computational Physics-II
Course Code: PHY-422 Part-II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2.	Write short answers of the following Questions:	
i.	Give simple example to find out zero elements of an array?	2
ii.	Write syntax with example for the following in MATLAB: (a) if-if() (b) gtext(), (c) polyint() and (d) diff()	8
iii.	Write MATLAB program code segment for the following: (a) to generate a matrix (5 x 6) having random entries and plot matrix rows (b) to give one example to 3D plot (c) to plot x vs y such as y= $3x^3 - 24x^2 + 8x + 42$ for [-15, 15] (d) to define two arrays and determine sum of two array values (e) to evaluate $\int_0^{\pi} \sin(x)dx$	10
Q.3.	Suppose A be a 3x3 matrix. Write MATLAB program which reads in random numbers as entries of the matrix A and calculate (i) sum and average of the all matrix elements, (ii) transpose of the matrix A (also plot the matrix), (iii) also check whether the Matrix A is an identity matrix? (iv) sort the matrix elements, (v) divide matrix rows by its row average.	10
Q.4. (a)	Write MATLAB program to calculate and print out factorial of a number taken from the user by using two methods. Implement your program using functions.	6+4
(b)	Suppose v, t and d are three linear arrays each of length is ten representing velocity, time and distance respectively. Write MATLAB program which reads in values for v and t & prints values of d. Also print values of total distance and average distance values	
Q.5. (a)	Write MATLAB program for the growth of Current in a simple RL-circuit using Euler's Method with initial conditions: r=10Ω, L=5H, initial time 0, time step 0.1, maximum time 2.5sec., initial current 0 and voltage v=4 volts. Print and plot current against time values. Draw estimate graphs with proper labels. How you can convert the same program for the decay of current in the circuit?	10
(b)	Write MATLAB program to plot time (t) against sin(t) and cos(t) curves on the same graph with different colors. Use t = [0 4π] Write MATLAB program to determine equivalent capacitor of n capacitors connected in parallel.	

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – 2020

Roll No. 019599

Paper: Computational Physics-II
Course Code: PHY-422 Part–II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2.	Write short answers of the following Questions:	
i.	Give simple example to find out standard deviation of an array?	2
ii.	Write syntax with example for the following in MATLAB: (a) for loop (b) while loop (c) if (d) sum ()	8
iii.	Write MATLAB program code segment for the following: (a) to generate and plot complex numbers (b) to find out integral of $x^3 + 4x^2 - 27x + 89$ (c) to find derivative of: $x^3 + 6x^2 - 21x + 290$ (d) to calculate the roots of the polynomials in (b) and (c) (e) to divide polynomial in (c) by the derivative of (b)	10
Q.3.	The acceleration of a spherical body moving in viscous medium is given by a = g – c v where c = 6 π η r / m with the conditions: **g = 9.8 m/sec^2, η=1.2, r = 0.2 m, m=1 kg, initial position x = 0, v=1 m/sec, h= 0.1 sec and tmax=2.5sec.** Write MATLAB program to plot and print time, position, velocity and acceleration values. Also draw estimate output graphs with proper curve labels, x & y labels and title. How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of a particle. Note: Plot estimate graph if any.	6+4
Q.4. (a)	Write MATLAB program for the simple harmonic motion(SHM) of a mass attached with a spring using Euler's method under the following conditions: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms. ω =0.01 s^{-1} and f_0=1.5N.). Calculate and print with proper labels the values of time against position, velocity and acceleration. How you can change the same program for the Forced H.M., Damped. H.M. The necessary equations are as follows: A = (-k x – b v + f_0 cos (wt))/ m, x = x + v h, v = v + a h, t = t + h. Also draw estimate output graphs with proper curve labels, x & y labels and title.	6+4
(b)	Write MATLAB program for a Full wave rectifier circuit. Note: Plot estimate graph if any.	
Q.5. (a)	Write a program with function to calculate, print and plot x and *f(x)* for $f(x) = x^3 + 6x^2 - 21x + 290$. Also find out sum, average, minimum of f(x) values for ten different values of x. Write MATLAB program to determine factorial of a number read from the user.	10
(b)	Write MATLAB program to determine equivalent resistance of n resistances connected in series.	

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Eighth Semester – 2019

Roll No. ....

Paper: Computational Physics-II
Course Code: PHY-422 Part–II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2.	Write short answers of the following Questions:	
i.	Give simple example to find zero index of values in an array?	2
ii.	Write syntax with example for the following in MATLAB: (a) ginput() (b) dsolve() (c) gtext() (d) input(),	8
iii.	Write MATLAB program code segment for the following: (a) to generate and plot 20 x 20 matrix of numbers (b) to give one example to find out cumulative sum of [2 5 8 7] (c) to find integral of a polynomial: $5x^2-2x+20$ (d) to calculate the roots of a polynomial (e) to multiply two polynomial and determine derivative	10
Q.3.	Suppose A be a 3x3 matrix. Write **MATLAB** program which reads in random numbers as entries of the matrix A and calculate (i) sum and average of the all matrix elements, (ii) transpose of the matrix A (also plot the matrix), (iii) also check whether the Matrix A is an identity matrix? (iv) Also sort the matrix elements, (v) divide matrix rows by its row average. Write MATLAB program for a half wave rectifier circuit.	6+4
Q.4. (a)	Write MATLAB program for the forced harmonic motion (FHM) of a mass attached with a spring using Euler's method under the following conditions: (g=9.8 m/s², initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms, ω=0.01 s^{-1} and f_0=1.5N.). Calculate and print with proper labels the values of time against position, velocity and acceleration. How you can change the same program for the Simple H.M., Damped. H.M. The necessary equations are as follows: A = (-k x – b v + f_o cos (wt))/ m, x = x + v h, v = v + a h, t = t + h, Also draw estimate output graphs with proper curve labels, x & y labels and title.	6+4
(b)	How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of a particle for 31 collisions. Also calculate the distance traced by the particle. Note: Plot estimate graph if any.	
Q.5. (a)	Write MATLAB program to calculate and print out factorial of a number taken from the user by using two methods. Implement your program using functions.	10
(b)	Calculate and print the series and sum of S, such that: $S = 77\sum_{k=1}^{20} k^3$ How you can improve the answer to evaluate $\int_0^{\pi} \sin(x)dx$	

UNIVERSITY OF THE PUNJAB

Eighth Semester – 2018

Examination: B.S. 4 Years Program

Roll No. ..12879..........

MAX. TIME: 2 Hrs. 45 Min.
MAX. MARKS: 50

PAPER: Computational Physics-II
Course Code: PHY-422 Part – II

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2.	Write short answers of the following Questions:	
i.	Give simple example to find nonzero index of an array?	2
ii.	Write syntax with example for the following in MATLAB: (a) If-else (b) std() (c) plot() (d) diff(),	8
iii.	Write MATLAB program code segment for the following: (a) to generate and plot complex numbers (b) to give one example to find out factorial of a number (c) to find derivative of a polynomial: $3x^3 + 7x^2 - 8x + 20$ (d) to calculate the circumference of a circle for radius =8 (e) to define two arrays and determine dot produce	10
Q.3.	Use matlab code to solve for [x,y,z] by two different methods. 2x+y+2z=17 2x-3y+4z=6 3y+z=6 Using rand(), create a matrix K (5 x 5). Then i) plot K, plot 2nd row and 5th rows. ii) find maximum of column 2 and iii) sort K. Also find the minimum and standard deviation value of K.	5+5
Q.4. (a)	The temperature measurement of containers A and B is given in the table. Write program to determine: (i) temperature of A less than that of B (ii) temp. of A equal to that of B with day (iii) temp. of A and B > 60 with days and (iv) find out the day when average obtained from A and B is minimum.	6+4
(b)	Write MATLAB code to plot 'x' against y(x) with supbplot such that: x=[-10 10], y = sin(nx). Use n=[1 4] and step size =1	
Q.5. (a)	The height h(t) and speed v(t) of a projectile launched with speed v_i at an angle A to the horizontal are given by: $h(t) = v_i\, t \sin A - 0.5\, g\, t^2$ $v(t) = (v_i^2 - 2\, v_i\, g\, t \sin A + g^2 t^2)^{1/2}$ the projectile will hit the ground when h(t) = 0 in time t_hit = 2 v_i sinA / g. If A = 40°, v_i =20 m/s and g=9.8 m/s^2 use MATLAB relational and logical operators to find the times when height is no less than 6 m and speed is simultaneously no greater than 16 m/s.	6+4
(b)	Write a program to calculate the equivalent resistance of n resistances in series. Take n from the user	

Table for Q.4 (a):

Day	1	2	3	4	5
A	60	56.2	63	64	58
B	54.6	56.2	64	60	59

UNIVERSITY OF THE PUNJAB

Eighth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Computational Physics-II — **TIME ALLOWED: 2 hrs. & 30 mins.**
Course Code: PHY-422 — **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2.	Write short answers of the following Questions:	
i.	Give simple example to find index of maximum value of the array?	2
ii.	Write syntax with example for the following in MATLAB: (a) for (b) sum() (c) rand() (d) dsolve(),	8
iii.	Write MATLAB program code segment for the following: (a) to generate a matrix (8 x 6) with all 1's as entries and plot matrix (b) to give one example to calculate polynomial derivative (c) to x vs y such as y= $7x^4 - 2x^3 - x + 1$ and x = [-15 15] (d) to define two arrays and determine sum of two array values (e) to evaluate $\int_0^{\pi} \exp(x)\sin(2\sqrt{x})dx$	10
Q.3.	If y = 4 x^5 Write MATLAB symbolic operations to calculate (i) first derivative w.r.t. x (ii) perform integration of your result. If y=f(x) = x^3- 6 x^2 +9 x +2, Write program to plot x vs y using x=[-1 5]. Find and plot marks at extremes. Find maximum of y. How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of a particle. Note: Plot estimate graph if any.	6+4
Q.4. (a) (b)	A mass m is suspended by three cables with tensions T_1, T_2 and T_3, which are related by the following equations: $T_1/\sqrt{35}$ -3$T_2\sqrt{34}$ +$T_3/\sqrt{42}$ =0 3$T_1/\sqrt{35}$ -4$T_3/\sqrt{42}$ =0 $T_1/\sqrt{35}$ -3$T_2\sqrt{34}$ +$T_3/\sqrt{42}$ =mg Write MATLAB program to solve for the tensions using three methods . Take mg=1. Write MATLAB program to determine equivalent capacitances of n resistances connected in series (Read capacitances from the user). Using rand(), create a matrix N (5 x 7). Then i) plot N, plot 3rd row and 5th rows. ii) find maximum of column 6 and iii) sort N. Also find the maximum and mean value of N.	4+6
Q.5. (a) (b)	Write MATLAB program to study the damped harmonic motion (DHM) of a mass attached with a spring using Euler's method under the following conditions: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms,). Print and plot values for time, position, velocity and acceleration. How you can change the same program for Simple harmonic motion? Write a program to a 3D plot in matlab. Take three variables as x,y and z.	7+3

Eighth Semester 2016

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Computational Physics-II

Course Code: PHY-422

TIME ALLOWED: 2 hrs. &

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2. Write short answers of the following Questions:

i. Give simple example to calcúlate average of an array? — 2

ii. Write syntax with example for the following in MATLAB: — 8
(a) if-else (b) diff() (c) rand() (d) max(),

iii. Write MATLAB program code segment for the following: — 10
(a) to generate a matrix of 10 by 10 numbers and plot matrix
(b) to give one example to calculate derivative
(c) to find roots of a polynomial: $x^4 + 7x^2 - 4x + 22$.
(d) to define two arrays and determine dot product of elements
(e) to evaluate $\int_0^{\pi} \exp(x)\sin(x)dx$

Q.3. The acceleration of a spherical body moving in viscous medium is given by $a = g - c\,v$ where $c = 6\pi\eta\ r/m$ with the conditions: $g = 9.8$ m/sec^2, η=1.2, r = 0.2 m, m=1 kg, initial position x = 0, v=1 m/sec, h= 0.1 sec and tmax=2.5sec. Write MATLAB program to plot and print time, position, velocity and acceleration values. Also draw estimate output graphs with proper curve labels, x & y labels and title. — 6+4

How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of a particle. Note: Plot estimate graph if any.

Q.4. (a) A parallel plate capacitor is constructed from two or more parallel conducting plates such that its capacitance C can be computed from the formula — 6+4

$$C = (n-1)\,\varepsilon A / d$$

Where n is number of plates, $\varepsilon = 8.85 \times 10^{-12}$ farad/meter is dielectric constant, A = 20 cm^2 is the area of each plate separated by distance d = 4 mm. Construct a table to print number of plates against capacitance values for a maximum of 10 plates.

(b) Create a matrix M in MATLAB expression such that

M =

$$\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 13 & 11 & 9 & 7 & 5 & 3 & 1 \\ 1 & 2 & 4 & 8 & 16 & 32 & 64 \end{matrix}$$

Also find the maximum and mean value of M.

Q.5. (a) The height h(t) and speed v(t) of a projectile launched with speed v_i at an angle A to the horizontal are given by: — 7+3

$$h(t) = v_i\, t \sin A - 0.5\, g\, t^2$$

$$v(t) = (v_i^2 - 2\, v_i\, g\, t \sin A + g^2 t^2)^{1/2}$$

the projectile will hit the ground when h(t) = 0 in time t_hit = 2 v_i sinA / g. If A = 40°, v_i =20 m/s and g=9.8 m/s^2 use MATLAB relational and logical operators to find the times when height is no less than 6 m and speed is simultaneously no greater than 16 m/s

(b) Write a program to calculate the equivalent resistance of n resistances in series. Take n from the user.

UNIVERSITY OF THE PUNJAB

Eighth Semester 2015

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Computational Physics-II

Course Code: PHY-422

TIME ALLOWED: 2 hrs. & 30 min

MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

NOTE: ATTEMPT ALL QUESTIONS. SUBJECTIVE

Q.2.	Write short answers of the following Questions:	
Repeat 2014 i.	Give simple example to calculate maximum of an array?	2
211, 203, 457 Palm ii.	Write syntax with example for the following in MATLAB: (a) for () loop (b) plot() (c) rand() (d) input(), → P#1210 Palm command P 11	8
iii.	Write MATLAB program code segment for the following: (a) to generate ten random numbers and plot bar graph registeo (b) to give one example to calculate derivative P#1605 Palm (c) to find roots of a polynomial: $x^3+4x^2-27x+89$ (d) to define two arrays and multiply both by elements (e) to evaluate $\int_0^{10} x^2 dx$	10 → P#29 registo. P#123 set②
Q.3.	(a) The temperature measurement of containers A and B is given in the table. Write program to determine: (i) temperature of A greater than that of B (ii) temperature of B equal to that of A with day (iii) temp. of B > 60 with days and (iv) find out the day when average temperature of A and B is maximum. Write (table below) (b) MATLAB program for conversions of length from m to km. → P# 49 set ③	8+2 P#46 palm
Q.4.	An electric resistance network has five resistances and two applied voltages. The current I_1, I_2 and I_3 are related by the following equations: $(R_1+R_4)I_1 - R_4I_2 = V_1$; $-R_4 I_1 + (R_2+R_4+R_5) I_2 - R_5 I_3 = 0$; $R_5 I_2 - (R_3+R_5)I_3 = V_2$. If R =[5 100 200 150 250] kΩ and V_1=100 & V_2=50V. Write MATLAB program to solve for the current values using three methods. Write MATLAB program to determine factorial of a number read from the user by using a user defined function. → P #12 set ④	7+3 P# 371 Palm
Q.5. Same Q4(A) 2016 Pu final	(a) Write MATLAB program for the Decay of Current in a simple RL-circuit using Euler's Method with initial conditions: r=10Ω, L=5H, initial time 0, time step 0.1, maximum time 2.5sec., initial current 12Amp and voltage v=4 volts. Print and plot current against time values. Draw estimate graphs with proper labels. How you can convert the same program for the growth of current in the circuit? (b) ~~Write a program with function to calculate~~ $f(x)$, given by: $f(x)=41x^3-14x^2+4x+21$. Registeo. Calculate, print and plot values for x and $f(x)$. Also find out sum, average, minimum of f(x) values for ten different values of x.	7+3 P# 82 set LC, RC circuit. P# 22, set ① P# 56 set ③

Day	1	2	3	4	5
A	60	56.2	63	64	58
B	54.6	56.2	64	60	59

Application of euler method.

$x = x + vh$

$v = v + ah$ P#59 set ①

$t = t + h$

Abdul Salam

UNIVERSITY OF THE PUNJAB

Eighth Semester 2014
Examination: B.S. 4 Years Programme

Roll No. 158

PAPER: Computational Physics-II
Course Code: PHY-422

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE

Q.2.		Write short answers of the following Questions:	
	i.	Give simple example to calculate an average of an array?	2
	ii.	Write syntax with example for the following in MATLAB: (a) while() loop (b) diff() (c) dsolve() (d) gtext(); P#77 Palm	8
P#222 Palm	iii.	Write MATLAB program code segment for the following: (a) to generate and plot complex numbers → P#30 register (b) to give one example to do partial derivative P# 605 Palm (c) to find roots of a polynomial: $x^3+6x^2-21x+290$ register (d) to calculate the area of a circle by taking the values of radius from the user P#58 set(3) P#31 (e) to evaluate $\int_0^{\pi}\sin(x)dx$ P#31 register	10
Q.3. set(3) P#51-52	(a)	The acceleration of a spherical body moving in viscous medium is given by a = g – c v where c = 6 π η r / m with the conditions: g = 9.8 m/sec^2, η=1.2, r = 0.2 m, m=1 kg, initial position x = 0, v=1 m/sec, h= 0.1 sec and tmax=2.5sec. Write MATLAB program to plot and print time, position, velocity and acceleration values. Also draw estimate output graphs with proper curve labels, x & y labels and title.	6+4 P#4 set(4)
	(b)	How randomly generated points can be used to show Brownian motion? Write MATLAB program to simulate Brownian motion of a particle. Note: Plot estimate graph if any.	
Q.4.	(a)	A mass m is suspended by three cables with tensions T_1, T_2 and T_3, which are related by the following equations: $T_1/\sqrt{35}-3T_2\sqrt{34}+T_3/\sqrt{42}=0$ $3T_1/\sqrt{35}-4T_3/\sqrt{42}=0$ $T_1/\sqrt{35}-3T_2\sqrt{34}+T_3/\sqrt{42}=mg$ Write MATLAB program to solve for the tensions using three methods. Take mg=1.	7+3
	(b)	Write MATLAB program to determine equivalent resistance of n resistances connected in series (Read resistances from the user).	
Q.5.	(a)	Write MATLAB program to study the damped harmonic motion (DHM) of a mass attached with a spring using Euler's method under the following conditions: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms,). Print and plot values for time, position, velocity and acceleration. How you can change the same program for Simple harmonic motion?.	7+3 P#160
	(b)	Write MATLAB code to plot 'x' against y(x) with supbplot such that: x=[-10 10], y = sin(nx). Use n=[1 4] and step size =1	Resistor mass & P#8 set(4)

Application –